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TECHNOLOGY NEEDS ASSESSMENT (TNA) REPORT

COMPANY:	Local Government Unit of Boac (LGU Boac)
ADDRESS:	Barangay Isok I, Boac, Marinduque

SCOPE OF ASSESSMENT*

The TNA covered the following areas:

1. Strategic Direction
 - a. Vision and mission
 - b. Plans and Objectives
 - c. Strategic alliances and current agreement
2. Management Aspect
 - a. Human resource management
 - b. Purchasing
 - c. Work environment
 - d. Corporate social responsibility
 - e. Occupational health and safety management
3. Technical Aspect
 - a. Operational and outsourcing practices
 - production system
 - production planning and control
 - production lay-out
 - work improvement
 - equipment management and maintenance
 - quality assurance system
 - outsourcing practices
 - b. Product and Process Performance and Improvement
 - reengineering and Research and Development
 - Procedures for continuous improvement
 - Product Quality standards
 - c. Environmental Management System
 - Waste management
4. Marketing Aspect
 - a. Marketing plan
 - b. Market outlets and number
 - c. Promotional strategies

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- d. Market competitors
- e. Packaging

5. Finance

- a. Cash flow and other related documents
- b. Sources of capital
- c. Accounting system

*** Scope of TNA is based on Technology Assessment Plan (TAP)**

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Summary of Assessment

Background

The Local Government Unit (LGU) of Boac, through its Information and Communication Technology (ICT) Section, serves as the primary implementing body for digital governance and ICT-driven services in the municipality. As the capital of Marinduque, LGU Boac has been actively advancing its digital transformation initiatives by adopting various computerized systems such as business permitting, taxation, accounting, and disaster response management. These efforts aim to enhance government efficiency, improve service delivery, and support data-driven decision-making.

Aligned with its vision of becoming a Smart and Sustainable Community, LGU Boac has invested in strengthening its ICT infrastructure, including the LGU Local Area Network (LAN), establishment of an ICT and Operations Center that supports centralized systems across barangays. In 2024, as part of its partnership with DOST-MIMAROPA under the Smart Sustainable Communities Program, the municipality implemented its Year 1 project focusing on the provision of a Hybrid Solar Energy System for the ICT Office and Command Center. This initiative was designed to address persistent power interruptions and ensure uninterrupted operations of critical information systems and emergency response services. Additionally, the LGU utilizes various information systems, including the Enhanced Tax Revenue Assessment and Collection System (ETRACS), Geographic Information System (GIS), and Human Resource Management System (HRMS).

Moreover, all barangays in the Municipality of Boac maintain their own individual internet subscriptions, which leads to increased and recurring expenses at the barangay level. This setup results in duplicated costs across the municipality and may not be the most cost-efficient approach, especially considering limited local

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resources. A more centralized or shared internet system could help reduce overall expenses while improving connectivity and service reliability for all barangays.

With regard to citizen-centric initiatives, reporting of accidents, incidents, and other related matters remains challenging due to the absence of a systematic and streamlined approach. As a result, information is not relayed promptly or efficiently, leading to delays in response and coordination among concerned offices. Establishing a centralized and structured reporting system would help ensure faster communication, improve response times, and enhance overall public service delivery.

Through this projects, LGU Boac not only improved the reliability of its ICT operations but also demonstrated its commitment to sustainable and innovative solutions. The collaboration with DOST-MIMAROPA highlights the LGU's proactive approach in integrating clean energy technologies to support digital governance, resilience, and long-term community development.

Methodology

The Technology Needs Assessment (TNA) was conducted through a systematic and participatory approach to identify, assess, and prioritize the technological requirements of the organization. The process began with a preparatory phase, where coordination with key stakeholders such as Information Communication Technology (ICT) Department and Science, Technology, and Innovation (STI) Committee was undertaken to define the scope, objectives, and expected outputs of the analysis, along with a review of existing documents, systems, and baseline data to establish the current technological context. This was followed by data gathering and assessment using a combination of methods such as key informant interviews, focus group discussions, surveys, and on-site observations to capture information on existing technologies, operational challenges, and user needs.

Subsequently, a technology gap analysis was performed to evaluate the difference between current systems and desired technological capabilities,

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focusing on aspects such as efficiency, reliability, and scalability. Based on this analysis, specific technology needs were identified and prioritized according to criteria such as urgency, impact on operations, cost-effectiveness, and alignment with organizational goals. The initial findings were then presented to stakeholders for validation to ensure accuracy and relevance, and to incorporate their feedback. Finally, appropriate recommendations were formulated, including proposed technological solutions, capacity-building interventions, and policy support, and these were documented in a comprehensive report that served as the basis for project planning, funding proposals, and implementation.

Summary of Findings

1. Strategic Direction

a. Vision and mission

Vision

A secure and resilient Boac: An empowered government, a cohesive society proud of its identity, an engaged citizenry.

Mission

We stand to promote a green economy and agriculture for sustainability and enhancing community resilience and safety. We are committed to adhere to good local governance in the form of an online platform, promoting countryside development for a circular economy approach to ensure a better, more sustainable and resilient socio-economic growth.

b. Plans and Objectives

The LGU of Boac plans to interconnect all barangays through a Municipal Local Area Network (MLAN) to ensure reliable and centralized internet connectivity. In addition, the LGU intends to develop a citizen-centric mobile application to streamline communication, reporting, and access to government services.

c. Strategic alliances and current agreements

The LGU Boac is currently engaged in collaborative initiatives with the Department of Science and Technology–MIMAROPA (DOST-MIMAROPA), other national government agencies (NGAs), and various non-government organizations (NGOs). One of its ongoing projects with DOST-MIMAROPA is the provision and installation of a Solar Energy System, aimed at supporting sustainable energy use.

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2. Management Aspect

a. Human resources

The LGU-Boac has an ICT Section that is responsible for managing and maintaining the municipality's digital systems, including the Local Area Network (LAN), information systems, and the Command Center operations, ensuring reliable connectivity and efficient delivery of ICT-enabled services.

b. Purchasing

The purchasing process of the LGU Boac is conducted in accordance with Republic Act No. 9184 or the Government Procurement Reform Act. The process begins with the identification of the end-user department's needs, followed by the preparation of the Project Procurement Management Plan (PPMP) and inclusion in the Annual Procurement Plan (APP). Procurement requests are then reviewed and approved by the Budget Office and the Bids and Awards Committee (BAC).

For procurement above the threshold, the BAC conducts public bidding, which includes the publication of bid opportunities, pre-bid conferences, submission and opening of bids, and post-qualification of the lowest calculated and responsive bidder. For small-value procurement, alternative methods such as shopping or negotiated procurement may be used, subject to legal requirements.

Once the winning supplier is determined, a Notice of Award and Notice to Proceed are issued, followed by contract signing. Delivery and inspection are conducted by the end-user and inspection team to ensure compliance with specifications. Payment is processed after complete delivery and acceptance of goods or services, supported by required supporting documents and approvals.

c. Business Ethics and Social Responsibility

The LGU Boac adheres to the principles of transparency, accountability, and integrity in all its transactions and decision-making processes. All procurement and service delivery activities are conducted in accordance with applicable government laws and regulations, particularly Republic Act No. 9184, to ensure fairness, competitiveness,

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and the proper use of public funds. Ethical standards are strictly observed to prevent corruption, conflict of interest, and other forms of misconduct.

In terms of social responsibility, the LGU prioritizes programs and projects that promote inclusive growth, environmental sustainability, and community welfare. It actively collaborates with national government agencies, non-government organizations, and local stakeholders to deliver services that address the needs of its constituents, particularly in education, health, infrastructure, and livelihood development. Through these efforts, the LGU of Boac aims to foster sustainable development and improve the overall quality of life of its residents.

3. Technical Aspect

a. Operational and Outsourcing Practices

Production system

The production system of the ICT Section and Command Center of the LGU Boac is an integrated system that ensures the efficient delivery of digital services, information management, and emergency response operations. It combines ICT infrastructure, software applications, network systems, and skilled personnel to support data processing, communication, and coordination across LGU offices. The ICT Section is responsible for maintaining and managing information systems, internet connectivity, databases, and technical support services, while also ensuring cybersecurity and system reliability. Meanwhile, the Command Center serves as the central hub for real-time monitoring and coordination during emergencies and routine operations, utilizing

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communication tools such as radio systems, surveillance cameras, and digital reporting platforms.

Production Planning and Control

The production and planning control of the ICT Section and Command Center of the LGU Boac involves the systematic organization, scheduling, monitoring, and evaluation of ICT-related operations and service delivery activities. Planning is guided by the LGU's annual development and investment priorities, where ICT projects, system upgrades, and command center enhancements are identified, scheduled, and aligned with available resources and procurement timelines. Production control ensures that all ICT services, including system maintenance, network operations, data management, and emergency response functions, are implemented efficiently and according to established standards and timelines. Continuous monitoring and coordination are conducted to track progress, address operational issues, and ensure minimal downtime. Through this structured planning and control system, the LGU ensures effective service delivery, optimal resource utilization, and timely response to the needs of its offices and constituents.

Production Layout

N/A

Work Study/Improvement

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The work study and improvement system of the ICT Section of the LGU Boac focuses on the continuous assessment and enhancement of workflows, processes, and ICT service delivery. It involves evaluating existing procedures, systems, and technologies to identify inefficiencies, delays, and areas that require optimization. Based on these assessments, improvements are implemented such as process streamlining, system upgrades, improved network configuration, and adoption of more efficient digital tools. The ICT Section also conducts regular monitoring and gathers feedback from end-users across LGU offices to ensure that services remain responsive, reliable, and aligned with operational needs. Through continuous work study and improvement initiatives, the LGU strengthens operational efficiency, enhances digital service delivery, and promotes effective use of ICT resources.

Equipment Management and Maintenance

The Equipment Management and Maintenance system of the ICT Section of the LGU Boac involves the proper inventory, utilization, monitoring, and upkeep of all ICT assets and related equipment. This includes computers, servers, networking devices, printers, and other peripheral devices used in the delivery of ICT services across LGU offices. Equipment management ensures that all assets are properly recorded, assigned, and utilized in accordance with office needs and standards, while maintenance activities focus on preventive and corrective measures to ensure optimal performance and minimize downtime. Regular inspections, software updates, hardware repairs, and system upgrades are conducted to maintain efficiency and extend the lifespan of equipment.

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Quality Assurance System

The Quality Assurance System of the LGU Boac ensures that all services, systems, and outputs consistently meet established standards of performance, reliability, and efficiency. It involves the implementation of policies, procedures, and monitoring mechanisms to evaluate the quality of government operations, including service delivery, program implementation, information systems, and administrative processes. Regular inspections, audits, and performance evaluations are conducted to identify gaps, ensure compliance with applicable laws and standards, and verify the effectiveness of government services. Feedback from stakeholders and constituents is also gathered and analyzed to support continuous improvement initiatives.

b. Product and Process Performance and Improvement

Performance Measures and Results – Process

The performance measures and results of the LGU Boac focus on evaluating the efficiency, effectiveness, and timeliness of its internal processes in delivering public services. Key performance indicators include compliance with prescribed procedures, processing time of transactions, resource utilization efficiency, and adherence to national government standards and regulations. Regular monitoring and evaluation are conducted to assess whether processes are being implemented as planned and to identify areas for improvement. Results are analyzed through reports, audits, and feedback mechanisms to determine process performance gaps and successes.

Performance Measures and Results – Product

The performance measures and results of the LGU Boac in terms of product focus on the quality, completeness, and effectiveness of the outputs delivered through its programs, projects, and services. These

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include infrastructure outputs, digital systems, public service facilities, and other tangible and intangible deliverables that address community needs. Evaluation is based on compliance with approved specifications, functionality, durability, and user satisfaction. Regular inspections, validation activities, and stakeholder feedback are conducted to ensure that outputs meet intended objectives and standards. The results of these assessments are used to improve planning, implementation, and resource allocation, ensuring that all LGU deliverables contribute to efficient governance and sustainable community development.

c. Environmental Management System

Waste Management

The Waste Management system of the LGU Boac focuses on the proper collection, segregation, transport, treatment, and disposal of solid waste in accordance with environmental laws and regulations. It is guided by Republic Act No. 9003 or the Ecological Solid Waste Management Act, which promotes waste reduction, segregation at source, recycling, and composting. The LGU implements regular garbage collection services and continuous information campaigns to encourage community participation in proper waste disposal practices. The Municipality of Boac operates a designated sanitary landfill located in Maybo, Boac, which serves as the primary disposal site for residual waste. In addition, the Material Recovery Facility (MRF) is also located in Maybo, Boac, where recyclable and biodegradable wastes are processed to support resource recovery and waste minimization efforts

4. Marketing Aspect

a. Marketing Plan

N/A

b. Market Outlets

N/A

c. Promotional Activities

The LGU Boac implements specific promotional activities to effectively disseminate information about its programs, projects, and services. These include regular updates on the official LGU social media platforms for announcements on government services,

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procurement activities, disaster advisories, and public programs. The LGU also conducts barangay information drives and community assemblies where officials and designated personnel present updates on projects, health programs, environmental initiatives, and livelihood opportunities. Public information materials such as posters, tarpaulins, and flyers are strategically placed in barangay halls and public areas to ensure wide coverage. In addition, radio announcements through local stations are utilized for urgent advisories and important public notices. The LGU also utilizes a LED wall located in the town proper to display real-time announcements, public advisories, and event promotions.

d. Market Competitors

N/A

5. Finance

a. *Source of capital/credit*

The LGU Boac sources its capital and credit primarily from national government allocations such as the Internal Revenue Allotment (IRA), grants, and funds from various national government agencies (NGAs) supporting local development programs and projects. Additional funding is also derived from local revenue sources, including taxes, fees, and charges collected by the municipality. For specific development initiatives, the LGU may also access financial assistance, subsidies, and counterpart funding from partner agencies such as the Department of Science and Technology (DOST) and other development partners. In some cases, financing support may be supplemented through loans or credit facilities from government financial institutions, subject to legal and budgetary requirements.

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b. Accounting System

The accounting system of the LGU Boac is designed in accordance with the New Government Accounting System (NGAS) prescribed by the Commission on Audit (COA). It ensures the proper recording, classification, summarization, and reporting of all financial transactions of the municipality. The system covers all sources of funds, including local revenues, national government allocations, grants, and other financial assistance, ensuring that all expenditures are properly documented and supported by legal and accounting requirements.

CONCLUSIONS:

Based on the interview and ocular inspection the TNA team concludes the following;

- While the LGU has established a Local Area Network (LAN), the current decentralized internet subscription model for individual barangays leads to duplicated costs and recurring expenses that strain local resources. This lack of a Municipal Local Area Network (MLAN) hinders cost-efficiency and unified service delivery.
- There is a significant absence of a streamlined approach for citizens to report accidents and incidents. This results in delayed communication and poor coordination among concerned offices, highlighting a critical need for a centralized, mobile-responsive reporting platform.
- The LGU Boac has a structured and well-coordinated governance system that supports efficient service delivery and public administration specially in ICT.

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- Operations are guided by established policies ensuring transparency, accountability, and compliance with government regulations.
- Strategic partnerships with NGAs, NGOs, and development partners strengthen the implementation of key programs and projects.
- The LGU effectively implements initiatives in infrastructure, ICT development
- Financial management systems, including accounting and procurement processes, support responsible and transparent use of public funds.
- Continuous efforts toward modernization and ICT integration improve efficiency and service delivery.
- Overall, the LGU Boac is positioned for sustainable and inclusive growth through continuous improvement and collaboration.

RECOMMENDATIONS:

The following are recommended by the TNA team:

- Strengthen the digital infrastructure of the LGU of Boac through the implementation of a Municipal Local Area Network (LAN) to ensure faster, more secure, and more reliable inter-barangay connectivity.
- Develop and deploy a citizen-centric app to provide constituents with a centralized platform for accessing government services, submitting requests, receiving announcements, and reporting concerns in real time.
- Enhance ICT capacity building programs for LGU personnel to ensure effective utilization and maintenance of new digital systems and platforms.

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- Improve data management and cybersecurity measures to safeguard government information and ensure system integrity.
- Expand ICT integration across all LGU departments to promote automation, efficiency, and interoperability of government services.
- Strengthen partnerships with NGAs, NGOs, and private sector stakeholders to support funding, technical assistance, and sustainability of ICT initiatives.
- Conduct regular system evaluations and user feedback assessments to continuously improve digital services and public engagement platforms.
- Allocate sufficient budget for ICT modernization projects, including hardware upgrades, software development, and network infrastructure expansion.

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